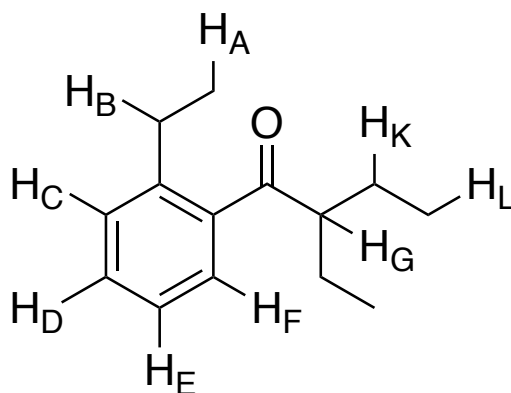


Chem 0345– Organic Lab

Activity 5: NMR Series 2

Extra Practice Problems

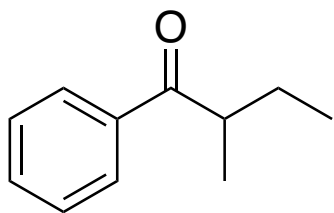
1. Consider the molecule below (Note that not all hydrogens are shown in the structure):



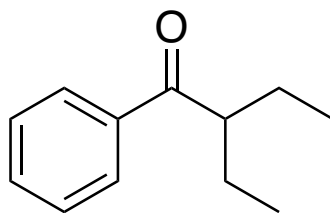
- Predict the expected splitting pattern of H_A : *Triplet (t)*
- Predict the expected splitting pattern of H_B : *quartet (q)*
- Predict the expected splitting pattern of H_C if $J_{CD} > J_{CE}$: *doublet of doublets (dd)*
- Predict the expected splitting pattern of H_C if $J_{CD} = J_{CE}$: *Triplet (t)*
- Predict the expected splitting pattern of H_D if $(J_{CD} = J_{ED}) > J_{FD}$: *Triplet of doublets (td)*
- Predict the expected splitting pattern of H_D if $J_{CD} > J_{ED} > J_{FD}$: *ddd*
- Predict the expected splitting pattern of H_G : *quintet*
- Predict the expected splitting pattern of H_K if $J_{GK} > J_{LK}$: *doublet of quartets*
- Predict the expected splitting pattern of H_K if $J_{GK} < J_{LK}$: *quartet of doublets*
- Predict the expected splitting pattern of H_K if $J_{GK} = J_{LK}$: *quintet*

2. How many distinct signals would you expect in the proton NMR of each of the molecules below?

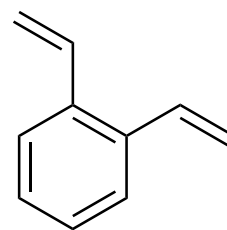
Note: Assume a high enough resolution that each signal is clearly distinguishable. Also assume that any N-H or O-H protons are observable in the NMR spectrum (if applicable).



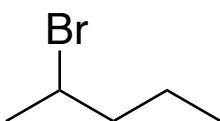
8



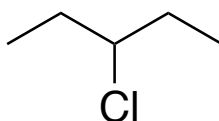
6



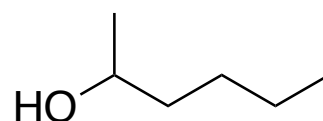
5



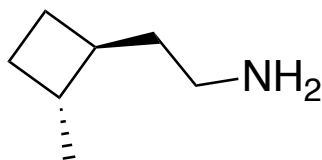
7



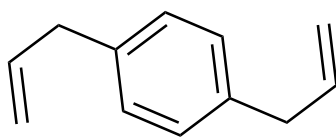
3



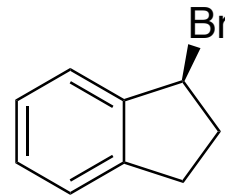
10



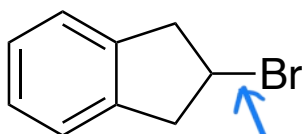
12



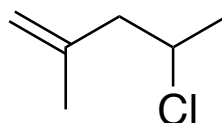
5



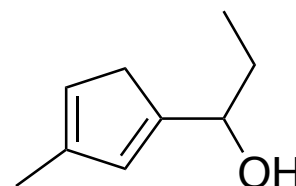
9



5



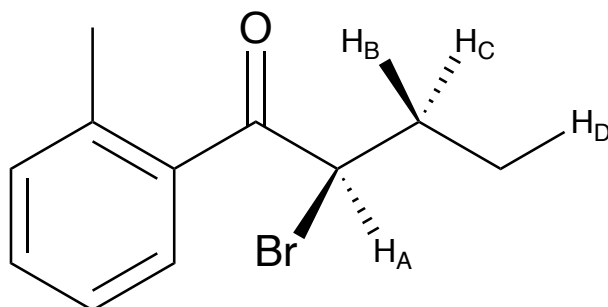
7



10

Remember that
this is on one
side of the ring.
How does this effect
the CH₂ groups of
the ring.

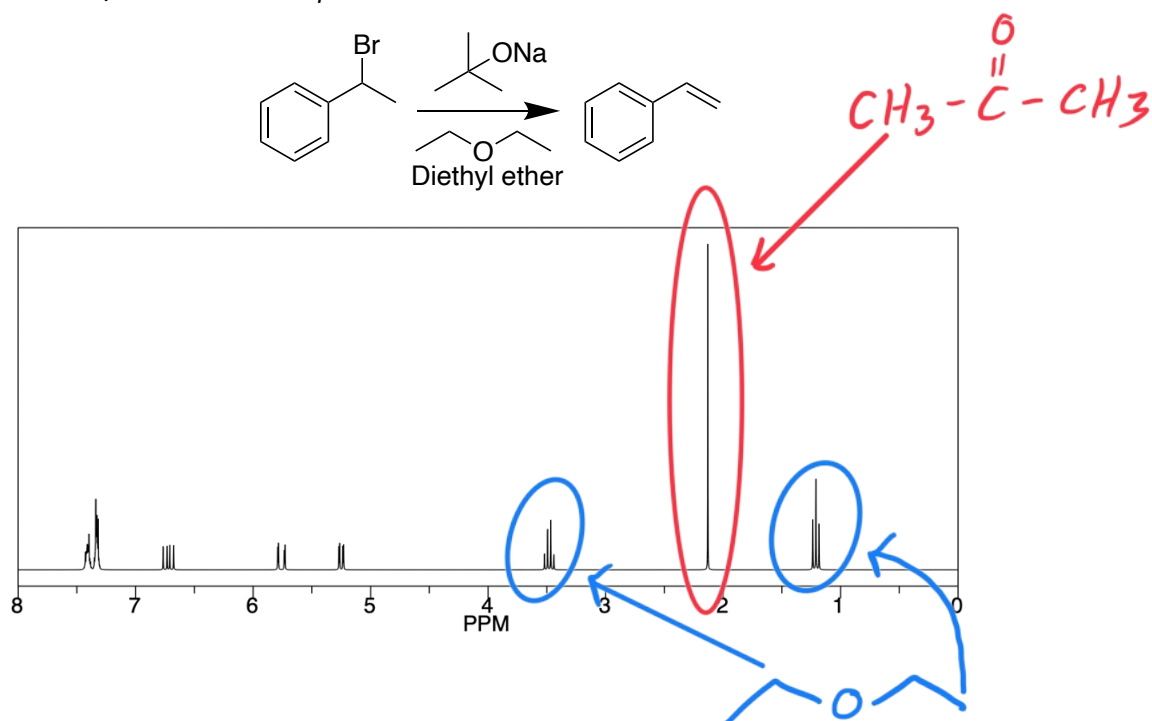
3. Consider the molecule below (Note: not all hydrogens are being shown. i.e. H_D is just one proton of the methyl group and two of the protons are not being shown):



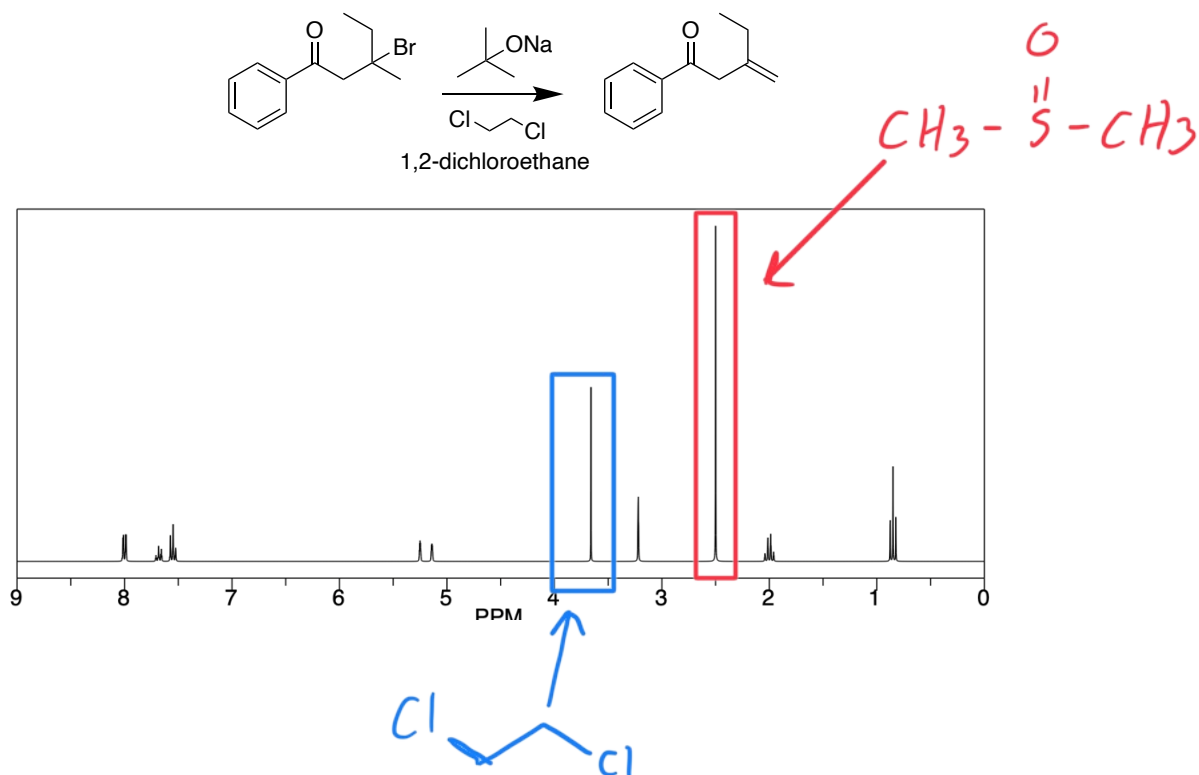
- a) Predict the expected splitting pattern of H_B if $J_{CB} > (J_{AB} = J_{DB})$: *doublet of quintets*
- b) Predict the expected splitting pattern of H_B if $J_{CB} > J_{AB} > J_{DB}$: *doublet of doublets of quartets*
- c) Predict the expected splitting pattern of H_B if all coupling constants are equal. *sextet*
- d) Predict the expected splitting pattern of H_A if $J_{BA} > J_{CA}$: *dd*
- e) Predict the expected splitting pattern of H_A if $J_{BA} = J_{CA}$: *t*
- f) Predict the expected splitting pattern of H_D if all coupling constants are equal: *t*
- g) Predict the expected splitting pattern of H_B if 2- and 3-bond couplings are observed and all 3-bond coupling constants are equal but $J_{2\text{-bond}} > J_{3\text{-bond}}$: *doublet of quintets*
- h) If the splitting pattern of H_D is observed to be a doublet of doublets, what can you conclude about the coupling constants?

$$J_{DB} \neq J_{DC}$$

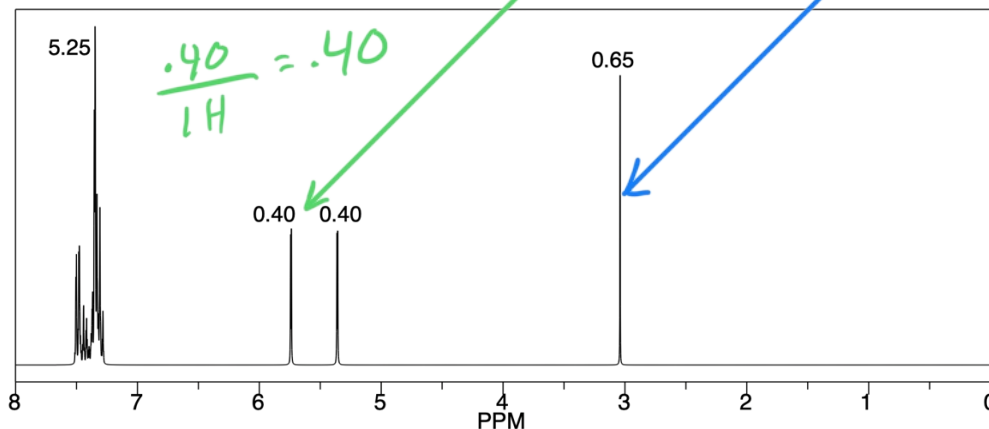
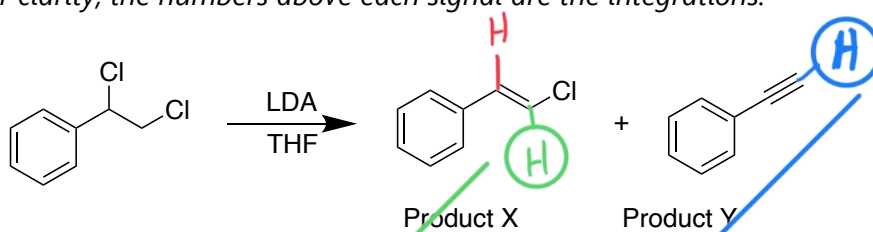
4. At the end of the following elimination reaction, an H-NMR spectrum of the crude product was acquired in $(\text{CD}_3)_2\text{CO}$. Identify all residual solvent signals from either the reaction solvent or the deuterated solvent used to prepare the NMR sample. *You may assume that all the starting reactants were consumed to form the desired product.*



5. At the end of the following elimination reaction, an H-NMR spectrum of the crude product was acquired in $(\text{CD}_3)_2\text{SO}$. Identify all residual solvent signals from either the reaction solvent or the deuterated solvent used to prepare the NMR sample. *You may assume that all the starting reactants were consumed to form the desired product.*



6. The following reaction resulted in a mixture of products (X and Y). Use the NMR spectrum to calculate the percentage of products X and Y. You may assume that no starting materials are present in the spectrum. For clarity, the numbers above each signal are the integrations.



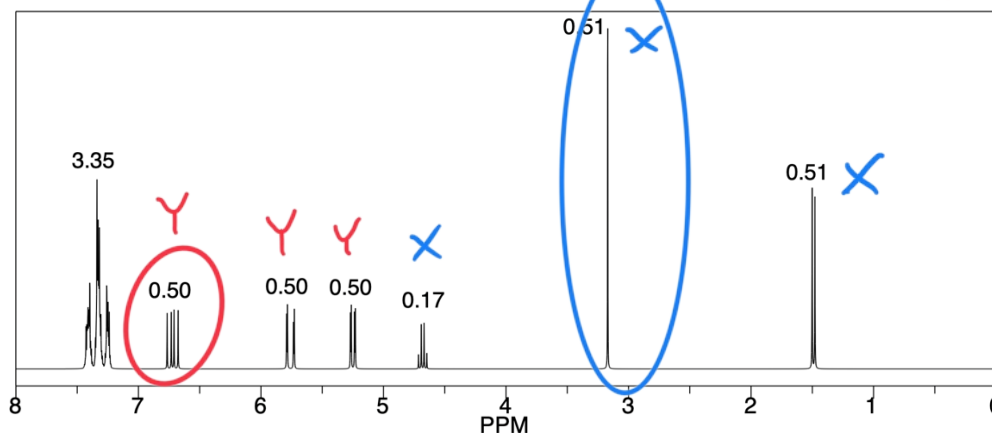
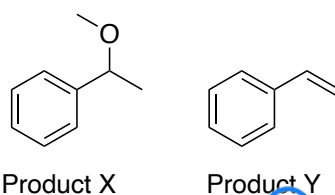
$$\% X = \frac{.40}{.40 + .65} = 39\%$$

$$\frac{.40}{1H} = .40$$

$$\frac{.65}{1H} = .65$$

$$\% Y = \frac{.65}{.65 + .40} = 62\%$$

7. Use the NMR spectrum to calculate the percentage of products X and Y. You may assume that no starting materials are present in the spectrum. For clarity, the numbers above each signal are the integrations.



$$\frac{.51}{3H} = .17$$

$$\% X = \frac{.17}{(.17 + .50)} \times 100 = 25\%$$

$$\frac{.50}{1H} = .50$$

$$\% Y = \frac{.50}{(.50 + .17)} \times 100 = 75\%$$

